MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 11: Complex integrals - examples

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Lecture outline

1. Examples

Example 6 (Lecture slides 9). Part 1.

Example

Evaluate $\oint_C \frac{2z+1}{z^3 - iz^2 + 6z} dz$ if C is the circle of radius $\frac{1}{3}$ about $3i$.

$$\oint_C \frac{2z+1}{z^3 - iz^2 + 6z} dz = \oint_C \frac{2z+1}{z(z^2 - iz + 6)} dz = \oint_C \frac{2z+1}{z(z-3i)(z+2i)} dz$$

$$\frac{2z+1}{z(z-3i)(z+2i)} = \frac{A}{z} + \frac{B}{z-3i} + \frac{C}{z+2i}$$

When we solve for A, B, and C, we get:

$A = \frac{1}{6}$

$B = \frac{-6}{15} i - \frac{1}{15}$

$C = \frac{2}{5} i - \frac{1}{10}$

Therefore

$$\oint_C \frac{2z+1}{z(z-3i)(z+2i)} dz = \oint_C \frac{1}{6z} dz + \oint_C \frac{\left(\frac{-6}{15} i - \frac{1}{15}\right)}{z-3i} dz + \oint_C \frac{\frac{2}{5} i - \frac{1}{10}}{z+2i} dz$$

Example 6. Part 2.

Example

Since $z = 0$ and $z = -2i$ lie outside C, then

$$\oint_C \frac{1}{6z} dz = 0$$

and

$$\oint_C \frac{\frac{2}{5}i - \frac{1}{10}}{z + 2i} dz = 0$$

Therefore, we have to solve only the second integral:

$$C : z(t) = 3i + \tfrac{1}{3}e^{it} \qquad 0 \leq t \leq 2\pi$$

$$z'(t) = \tfrac{1}{3}ie^{it}$$

So we get,

$$\int_0^{2\pi} \frac{(\frac{-6}{15}i - \frac{1}{15})}{\frac{1}{3}e^{it}} \frac{1}{3} ie^{it} dt = [i(\frac{-6}{15}i - \frac{1}{15})t]_0^{2\pi} = 2\pi i(-\frac{6}{15}i - \frac{1}{15})$$

Example 1. Part 1.

Example

Evaluate

$$\oint_C [z - Re\, z]\, dz,$$

where C is the circle $|z| = 2$.

$$\oint_C [z - Re\, z]\, dz = \oint_C z dz - \oint_C Re\, z\, dz$$

Since z is analytic everywhere on and inside C, then by Cauchy's theorem,

$$\oint_C z dz = 0$$

.

Example 1. Part 2.

Example

$Re(z)$ is analytic nowhere (by C-R - slide No 23 in lecture slides 4, and below) in the complex plane by, so then, we must parameterize:

$$C : z(t) = 2\cos t + 2i\sin t \qquad 0 \le t \le 2\pi$$
$$z'(t) = 2ie^{it}$$

So the integral becomes,

$$-\int_0^{2\pi} 2\cos t[2i[\cos t + i\sin t]]dt = -\int_0^{2\pi} [4i\cos^2 t + 4i\sin t\cos t]dt$$
$$= -2it + i\sin 2t + 2i\sin^2 t]_0^{2\pi} = -4\pi i.$$

Example

$f(z) = Re[z] = x$. here $u = x$ and $v = 0$ so that $u_x = 1, u_y = v_x = v_y = 0$. Thus the C-R Equations are satisfied nowhere so that $f(z) = x$ is differentiable nowhere.

Example 2. Part 1.

Example

Evaluate:

$$\oint_\Gamma f(z)\,dz$$

in each case below. Γ is oriented counterclockwise.

a. $f(z) = \dfrac{1}{(z-2i)^3}$ and Γ is given by $|z-2i| = 2$;

Since $2i$ is inside the circle, then we cannot use Cauchy's theorem.

So then, $\Gamma : z(t) = 2i + 2e^{it} \qquad 0 \le t \le 2\pi$

$$z'(t) = 2ie^{it}$$

$$\int_0^{2\pi} \frac{2ie^{it}dt}{8e^{3it}} = \int_0^{2\pi} \frac{1}{4} ie^{-2it} dt = -\frac{1}{8}e^{-2it}\Big]_0^{2\pi} = -\frac{1}{8}e^{-4\pi i} + \frac{1}{8} = 0.$$

Example 2. Part 2.

Example

Evaluate:

$$\oint_{\Gamma} f(z)\, dz$$

in each case below. Γ is oriented counterclockwise.

b. $f(z) = \bar{z}$ and Γ is the unit circle about the origin.

Since $\bar{z}$ is analytic nowhere, we cannot use Cauchy's theorem.

$\Gamma : z(t) = e^{it} \qquad 0 \leq t \leq 2\pi$

$z'(t) = ie^{it}$

$$\int_0^{2\pi} e^{-it}[ie^{it}]dt = \int_0^{2\pi} idt = it]_0^{2\pi} = 2\pi i.$$

Example 3. Part 1.

Example

Use partial fractions to evaluate

$$\oint_\Gamma \frac{1}{z^2+8z}\,dz$$

where Γ is the unit circle about the origin, oriented counterclockwise.

$$\oint_\Gamma \frac{1}{z^2+8z}\,dz = \oint_\Gamma \frac{1}{z(z+8)}\,dz$$

$$\frac{1}{z(z+8)} = \frac{A}{z} + \frac{B}{z+8}$$

Then from this we get,

$A = \frac{1}{8}$
$B = -\frac{1}{8}$

Example 3. Part 2.

Example

$$\oint_\Gamma \frac{1}{z(z+8)}dz = \oint_\Gamma \frac{dz}{8z} - \oint_\Gamma \frac{dz}{8(z+8)}$$

Since $z = -8$ is outside of Γ, by Cauchy's theorem,

$$-\oint_\Gamma \frac{dz}{8(z+8)} = 0.$$

Now $\Gamma : z(t) = e^{it} \qquad 0 \leq t \leq 2\pi$
$z'(t) = ie^{it}$

So,

$$\oint_\Gamma \frac{dz}{8z} = \int_0^{2\pi} \frac{ie^{it}}{8e^{it}}dt = \int_0^{2\pi} \frac{i}{8}dt = \frac{it}{8}\Big]_0^{2\pi} = \frac{\pi i}{4}$$

Example 4. Part 1.

Example

Evaluate

$$\oint_\Gamma \frac{e^z \sin\left(z^2\right)}{z-2} dz$$

over any closed path Γ that does not pass through 2.

We have 2 cases:

Case 1. Γ does not enclose 2.

So then, $\dfrac{e^z \sin\left(z^2\right)}{z-2}$ is differentiable on Γ and in its interior.

So by the Cauchy Integral theorem,

$$\oint_\Gamma \frac{e^z \sin\left(z^2\right)}{z-2} dz = 0$$

Example 4. Part 2.

Example

Case 2. Γ encloses 2. Look at this in light of the Cauchy Integral Formula.

Let $f(z) = e^z \sin\left(z^2\right)$

So,

$$f(2) = \frac{1}{2\pi i} \oint_\Gamma \frac{e^z \sin\left(z^2\right)}{z-2} dz$$

So then,

$$\oint_\Gamma \frac{e^z \sin\left(z^2\right)}{z-2} dz = (2\pi i) e^2 \sin(4)$$

Example 5. Part 1.

Example

Evaluate

$$\oint_\Gamma \frac{4e^{z^2}}{(z-i)^3}\,dz,$$

where Γ is any closed path in the plane not passing through i.

We have 2 cases:

Case 1. Γ does not enclose i.

So then, $\dfrac{4e^{z^2}}{(z-i)^3}$ is differentiable on Γ and in its interior.

Then by the Cauchy Integral theorem,

$$\oint_\Gamma \frac{4e^{z^2}}{(z-i)^3}\,dz = 0$$

Example 5. Part 2.

Example

Case 2. Γ encloses i. Look at this in light of the Cauchy Integral formula for higher order derivatives with $n = 2$.

Let $f(z) = 4e^{z^2}$

$f'(z) = 4e^{z^2}(2z) = 8ze^{z^2}$.

$f''(z) = 8e^{z^2} + 16z^2e^{z^2}$

Then,

$$f''(i) = \frac{2!}{2\pi i}\oint_\Gamma \frac{4e^{z^2}}{(z-i)^3}dz$$

So then,

$$\oint_\Gamma \frac{4e^{z^2}}{(z-i)^3}dz = \pi i[8e^{-1} - 16e^{-1}] = -8\pi ie^{-1}$$

Example 6. Part 1.

Example

Evaluate

$$\oint_\Gamma \frac{2\sin\left(z^3\right)}{(z-1)^4}\,dz,$$

where Γ is any closed path in the plane not passing through 1.

We have 2 cases:

Case 1. Γ does not enclose 1.

So then,

$$\frac{2\sin\left(z^3\right)}{(z-1)^4}$$

is differentiable on Γ and in its interior.

Therefore, by the Cauchy Integral theorem,

$$\oint_\Gamma \frac{2\sin\left(z^3\right)}{(z-1)^4}\,dz = 0.$$

Example 6. Part 2.

Example

Case 2. Γ encloses 1.

Then, let $f(z) = 2\sin(z^3)$ and apply the Cauchy integral formula for higher derivatives with $n = 3$.

$f'(z) = 6z^2\cos(z^3)$

$f''(z) = 12z\cos\left(z^3\right) - 18z^4\sin(z^3)$

$f'''(z) = 12\cos\left(z^3\right) - 36z^3\sin(z^3) - 72z^3\sin(z^3) - 54z^6\cos\left(z^3\right)$

So,

$$f'''(1) = \frac{3!}{2\pi i}\oint_\Gamma \frac{2\sin\left(z^3\right)}{(z-1)^4}dz.$$

Finally,

$$\oint_\Gamma \frac{2\sin\left(z^3\right)}{(z-1)^4}dz = \frac{\pi i}{3}[12\cos(1) - 36\sin(1) - 72\sin(1) - 54\cos(1)]$$

$$= \frac{\pi i}{3}[-42\cos(1) - 108\sin(1)] = -14\pi i\cos(1) - 36\pi i\sin(1).$$

Example 7.

Example

Prove that every non-constant polynomial $p(z)$ with complex coefficients of degree $n \geq 1$ has exactly n roots. (Note: some or all of these roots may be repeated.)

We will prove this by induction on n.

$n = 1$:
$p(z) = Az + B$ where A and B are complex constants and $A \neq 0$.
Then, $Az + B = 0$ implies that $z = \frac{-B}{A}$.
So there is one root.

Suppose any polynomial of degree $n \geq 1$ has exactly n roots.
Consider the polynomial $p(z)$ of degree $n + 1$.
By the fundamental theorem of algebra, $p(z)$ has at least 1 root.
Call this root $z = a$. Then, $p(z) = (z - a)q(z)$ where $q(z)$ is of degree n.
By the inductive hypothesis, $q(z)$ has exactly n roots. So then, $p(z)$ has exactly $n + 1$ roots.

Example 8. Part 1.

Example

Prove Morera's Theorem: If $f(z)$ is continuous in a simply-connected region R, and if

$$\oint_C f(z)\,dz = 0$$

around every simple closed curve C in R, then $f(z)$ is analytic in R.
(Note: This is a converse of Cauchy's Theorem.)

Suppose that $f(z)$ is continuous in a simply-connected region R and

$$\oint_C f(z)dz = 0$$

around every simple closed curve C in R.
Then, this is equivalent to the fact that $f(z)$ has an antiderivative $F(z)$ in R. Since $f(z) = F'(z)$, then $F(z)$ must be analytic in R. Therefore, $F(z)$ has derivatives of all order. This implies then, that $f(z)$ must be analytic in R.

Thank you